

CAR foundations and canonical states

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Scope

Finite matrix stages enter the CAR completion with distinct root-state and normalized-trace roles. Pointwise root-compression estimates support later residual-space arguments, while simplicity and compact-image exclusion are separate conclusions.

12 direct Cards + 0 reused prerequisites = 12 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key construction: Constructing the normalized trace on the completed CAR algebra
- Key result: Root compression becomes scalar in norm

[Read this route with prerequisites](#)

Detailed route conditions

The finite matrix stages $A_n = M_{2^n}(\mathbb{C})$ enter their norm completion A through ι_n . The root state ω and normalized trace τ have different roles: $\omega(\iota_n(a)) = a_{00}$, whereas $\tau(\iota_n(a)) = 2^{-n} \text{Tr}(a)$. The root projections are $f_n = \iota_n(E_{00})$. Here E_{00} belongs to A_n and $f_n = e_{00}^{(n)}$ is its image in A ; that image need not have rank one in a representation of A .

The compression estimate $\|f_n x f_n - \omega(x) f_n\| \rightarrow 0$ is pointwise in the fixed element x . It supplies the later residual-space arguments. The simplicity proof uses a sufficiently late fixed projection; it does not replace this norm estimate by uniform convergence. Infinite dimension and the absence of nonzero compact images in irreducible representations are separate conclusions. Their proofs remain in the individual Cards.

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (2 Cards)

Level 0

The CAR algebra as a completion of finite matrix stages

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

`MathlibAnnex.CStarAlgebra.CAR.Limit`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The product-vector state on the completed CAR algebra · A finite-row average centralizes its matrix stage · The decreasing root projections in the CAR completion · Constructing the normalized trace on the completed CAR algebra · The completed CAR algebra is infinite-dimensional

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Level 0

The normalized trace on a finite CAR stage

The normalized matrix trace provides a positive tracial functional compatible with the CAR inclusions.

`MathlibAnnex.CStarAlgebra.CAR.stageTrace`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: Constructing the normalized trace on the completed CAR algebra

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Level 1 (5 Cards)

Level 1

Constructing the normalized trace on the completed CAR algebra

Carries compatible bounded matrix traces through the algebraic limit, normed union and completion.

`MathlibAnnex.CStarAlgebra.CAR.trace`

Immediate Card prerequisites: The normalized trace on a finite CAR stage · The CAR algebra as a completion of finite matrix stages

Used by in this scope: Normalization and the trace identity determine the CAR trace

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Level 1

The product-vector state on the completed CAR algebra

Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion.

`MathlibAnnex.CStarAlgebra.CAR.rootState`

Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages

Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state

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Level 1 The completed CAR algebra is infinite-dimensional Rules out finite dimension by retaining matrix subalgebras of unbounded dimension. MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: An irreducible CAR representation has no nonzero compact image
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Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm
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Level 1 A finite-row average centralizes its matrix stage Turns an arbitrary element of the completed CAR algebra into one commuting with a prescribed finite matrix stage. MathlibAnnex.CStarAlgebra.CAR.ofStage_commute_rowAverage
Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: None in this selected scope
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Level 2 (3 Cards)

Level 2

Root compression becomes scalar in norm

Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra.

`MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion

Used by in this scope: Simplicity of the completed CAR algebra

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Level 2

Purity of the completed root state

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

`MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState`

Immediate Card prerequisites: The product-vector state on the completed CAR algebra

Used by in this scope: None in this selected scope

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Level 2

Normalization and the trace identity determine the CAR trace

Proves uniqueness without assuming positivity of the competing continuous functional.

`MathlibAnnex.CStarAlgebra.CAR.eq_trace_of_apply_one_of_mul_comm`

Immediate Card prerequisites: Constructing the normalized trace on the completed CAR algebra

Used by in this scope: None in this selected scope

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Level 3 (1 Card)

Level 3

Simplicity of the completed CAR algebra

Every nonzero closed two-sided ideal of the CAR algebra contains the identity.

`MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit`

Immediate Card prerequisites: Root compression becomes scalar in norm

Used by in this scope: An irreducible CAR representation has no nonzero compact image

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Level 4 (1 Card)

Level 4

An irreducible CAR representation has no nonzero compact image

Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements.

`MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image`

Immediate Card prerequisites: Simplicity of the completed CAR algebra · The completed CAR algebra is infinite-dimensional

Used by in this scope: None in this selected scope

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